

EZRA JOHNSON

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EDUCATION

Bachelor of Computer Science, California Institute of Technology

December 2025

FORTHCOMING PUBLICATIONS

- **E. Johnson**, A. Lechowicz, M. Hajiesmaili. *Signal-Aware Workload Shifting Algorithms with Uncertainty-Quantified Predictors*, ACM International Conference on Future Energy Systems (e-Energy), 2026.

RESEARCH EXPERIENCE

NSF REU Researcher

January 2026 - April 2026

University of Massachusetts Amherst

Remote

- Reviewed multi-armed bandit literature and designed study of new variant of multi-agent multi-armed bandit evaluated using metrics of fairness.
- Proved regret bounds and Nash social welfare of the algorithms developed.
- Engineered simulation code to empirically test the distributed algorithms developed.

NSF REU Summer Researcher

June 2025 - August 2025

University of Massachusetts Amherst

Amherst, MA

- Designed and proved theoretical bounds for learning-augmented algorithms in uncertainty-quantified settings.
- Scripted simulation code to test our algorithms on both a carbon forecasting dataset and an electricity market price dataset.
- Authored a publication to appear in e-Energy. Presented work at a summer poster session.

Undergraduate Researcher

March 2025 - June 2025

Caltech

Pasadena, CA

- Proposed and executed an independent project on the equity of cascading power failures in the California energy grid under Professor Adam Wierman.
- Presented findings at the Caltech Computing and Mathematical Sciences department poster session.

Undergraduate Researcher

January 2024 - March 2025

Caltech

Pasadena, CA

- Collaborated with Professor R. Michael Alvarez to design and carry out a project on identifying climate change misinformation in multi-modal video content using machine learning in the Alvarez Political Science group.
- Led the project through IRB review, collected and analyzed thousands of relevant videos, and evaluated models on their misinformation detection performance.

INDUSTRY EXPERIENCE

Software Engineer Intern

June 2024 - Sept 2024

Stripe

New York, NY

- Implemented personalized suggestions for the Universal Help widget to create a unique experience for support users on the Merchant Support Experience team using Ruby and TypeScript.
- Designed the personalization modules and wrote a proposal that went through design review.
- Generated machine learning predictions for over 170,000 support cases producing valuable data to reduce support agent costs.

Software Engineer Intern
Stripe

June 2023 - Sept 2023
New York, NY

- Adapted existing API and database models to provide essential transaction information to clients using Apache Kafka and Ruby.
- Updated the Stripe Dashboard in React to show my changes to the models.
- Saved company tens of man-hours per week and followed through on a highly requested client feature.

Software Engineer Intern
Meta (Facebook)

June 2022 - Sept 2022
New York, NY

- Developed the backend of a new feature for Instagram Stories as a server engineer using Django.
- Onboarded full-time engineers onto the project to take over after my internship duration.
- Presented on my work in an org-wide event.

STEP Intern
Google

June 2021 - Sept 2021
Remote

- Developed an internal web application using JavaScript, Google Closure Templates, a microservice framework, and more on the Google Lens in Photos Team.
- Collaborated closely with another intern to develop the product, and conducted code reviews.
- Utilized the Mercurial version management system.

AWARDS

- Awarded an NSF REU through the Computing for an Equitable Energy Transition at the University of Massachusetts Amherst.
- Earned a Southern California Intercollegiate Athletic Conference All-Conference Discus Award.

SIDE PROJECTS

Machine Learning for Medical Image Diagnosis. Developed a machine learning pipeline to produce a model that could identify 9 diseases from the CheXpert chest X-ray dataset in a small team. Utilized primarily Python and PyTorch.

Mobile Robot. Constructed a mobile robot capable of localization, mapping, and planning around obstacles in a team of two. Utilized Python and ROS2.

TECHNICAL SKILLS

Programming Languages
Tools

Python, Ruby, Java, C, OCaml, JavaScript, HTML, CSS
Git, Mercurial, NumPy, React, NextJS, PyTorch